

Agricultural Data Preprocessing and Cleaning Workflow

Course Assignment: Data Collection and Preparation for Machine Learning

Topic: End-to-End Simulation of Collecting, Preprocessing, and Cleaning Agricultural Datasets

1. Introduction & Executive Summary

In modern agribusiness and precision agriculture, machine learning models play a critical role in yield estimation, crop disease forecasting, and market price stabilization. However, real-world agricultural data is inherently complex, noisy, and distributed across multiple domain spaces—ranging from satellite earth observations to local wholesale market arrivals. This assignment documents a realistic simulation of acquiring three distinct public datasets, evaluating their structures, and building a robust data cleaning and preprocessing pipeline tailored for machine learning readiness.

2. Data Source Exploration & Relevance

To capture the multi-faceted nature of crop productivity and agribusiness economics, I selected three publicly available datasets spanning crop yield statistics, environmental weather metrics, and market commodity prices.

Dataset Name	Source / Platform	Data Type & Structure	Agribusiness Relevance & ML Utility
Global Crop Yield Statistics	Food and Agriculture Organization (FAOSTAT)	Tabular (CSV), Annual aggregates per country/region	Establishes ground-truth target variables (e.g., yield in hectograms per

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			hectare) for long-term historical trends and country-level yield predictions.
Surface Weather & Agro-Climatology	NASA POWER Project	Time-Series / Geospatial (CSV, JSON API), Daily observations	Provides primary environmental features (rainfall, min/max temperatures, solar radiation, humidity) required to model crop growth conditions and stress factors.
Daily Commodity Market Prices	Agmarknet / Open Govt Data (OGD) / USDA AMS	Tabular (CSV/API), Daily market-level arrivals & prices	Enables supply chain forecasting, price volatility modeling, and economic analysis by tracking arrival volumes along with minimum, maximum, and modal prices.

Detailed Dataset Breakdown:

- **FAOSTAT Crop Production:** Contains detailed records on crop type, area harvested, production volume, and yield across global regions. Essential for defining ground-truth targets in predictive yield models.

- **NASA POWER Meteorological Data:** Provides solar radiation, surface temperature, humidity, and precipitation data mapped to specific latitude/longitude points, providing key abiotic stress parameters for crop growth modeling.
- **Agmarknet Market Arrivals:** Records daily commodity arrivals across regional agricultural markets (mandis), serving as a crucial baseline for economic supply chain modeling and price forecasting.

3. Preprocessing & Feature Engineering Overview

Because raw data originates from diverse sources at varying temporal and spatial granularities, direct model ingestion is impossible without systematic preprocessing.

Temporal Alignment & Aggregation

Weather data is recorded on a daily basis, whereas crop yield targets are reported annually. To bridge this gap, daily weather records are temporally aggregated into growing-season metrics rather than simple annual averages. Key aggregations include:

- **Total Accumulated Rainfall:** Total precipitation during critical phenological stages (e.g., vegetative vs. flowering phases).
- **Thermal Stress Hours:** Count of days where temperatures exceeded critical crop thresholds (e.g., $>35^{\circ}\text{C}$).

Domain-Specific Feature Extraction

- **Growing Degree Days (GDD):** Calculated as $\text{GDD} = ((T_{\text{max}} + T_{\text{min}}) / 2) - T_{\text{base}}$ to quantify heat accumulation relevant to crop maturation.
- **Price Volatility Index:** Rolling 7-day and 30-day standard deviations of modal market prices to capture market instability for economic risk modeling.

Feature Scaling & Transformation

Agricultural metrics operate across vastly different scales—for instance, rainfall (mm), solar exposure (MJ/m^2), and price per quintal. To ensure model convergence and prevent feature dominance:

- **Min-Max Scaling:** Applied to bounded features for distance-based algorithms (e.g., K-Nearest Neighbors, Neural Networks).
- **Standard Scaling (Z-Score Normalization):** Applied to Gaussian-like features prior to feeding into linear regression models or SVMs.

4. Data Cleaning Simulation

Data quality issues such as sensor dropouts, holiday market gaps, and recording typos are prevalent in agricultural datasets. Below is the structured cleaning methodology implemented for this pipeline.

A. Missing Value Handling Strategy

Data Feature	Nature of Missingness	Imputation Technique & Justification
Daily Weather (Precipitation/Temp)	Short gaps (1-3 days) from sensor outages	Linear / Spline Interpolation: Preserves continuous environmental trends without creating sudden artificial spikes.
Weather Metrics	Extended gaps (>7 days)	Historical Contextual Median: Replaced with the multi-year monthly spatial median for that specific coordinate.
Market Modal Prices	Market closures, weekends, and holidays	Forward Fill (FFill): Carries over the last reported market price, assuming price stability over short non-trading intervals.
Crop Yield Target	Missing historical records for an entire year	Row Exclusion: Rows missing target labels are dropped to prevent introducing synthetic bias into the target feature.

B. Outlier Detection & Treatment

In agriculture, extreme values can represent either physical recording errors (e.g., a weather station outputting 100°C) or actual extreme events (e.g., catastrophic drought or severe flood). The distinction dictates the treatment:

- **Detection Method:** Interquartile Range (IQR) boundary evaluation, where values lying beyond $Q1 - 1.5 * IQR$ or $Q3 + 1.5 * IQR$ are flagged.
- **Handling Strategy:** Sensor errors and physically impossible entries (e.g., negative prices, extreme temperatures) are capped at the 5th and 95th percentiles (**Winsorization**). Verified climate extremes are preserved to allow the model to learn climate anomaly responses.

C. Duplicate Detection

Duplicate records frequently occur when aggregating market records across multiple sub-agencies. Duplicates are identified using a composite key: [Record_Date + Spatial_Location + Commodity_Name]. When detected, the primary verified entry is retained and subsequent duplicates are purged and logged.

5. Python Implementation Simulation

The following Python script illustrates the complete data cleaning and preprocessing pipeline, encapsulating the deduplication, missing value imputation, outlier winsorization, and normalization steps.

```
import pandas as pd
import numpy as np

def clean_and_preprocess_agri_data(df):
    """
    Simulates end-to-end data cleaning and preprocessing for agricultural datasets.
    """
    print("Initial Data Shape:", df.shape)

    # Step 1: Remove Duplicate Entries based on composite primary key
    composite_keys = ['Date', 'Region_ID', 'Crop_Type']
    df = df.drop_duplicates(subset=composite_keys, keep='first')
    print("Shape after Deduplication:", df.shape)

    # Step 2: Handle Missing Values
    # Linear interpolation for continuous weather variables (short gaps)
    if 'Precipitation_mm' in df.columns:
        df['Precipitation_mm'] =
df['Precipitation_mm'].interpolate(method='linear')
```

```

    # Forward-fill missing daily market prices (up to 3 consecutive days)
    if 'Modal_Price' in df.columns:
        df['Modal_Price'] = df.groupby('Region_ID')['Modal_Price'].transform(
            lambda x: x.ffill(limit=3)
        )
        # Fill remaining long gaps with regional median
        df['Modal_Price'] = df['Modal_Price'].fillna(
            df.groupby('Region_ID')['Modal_Price'].transform('median')
        )

# Step 3: Outlier Detection and Winsorization (Capping)
if 'Modal_Price' in df.columns:
    Q1 = df['Modal_Price'].quantile(0.25)
    Q3 = df['Modal_Price'].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Cap extreme values to acceptable boundary limits
    df['Modal_Price_Capped'] = np.where(
        df['Modal_Price'] > upper_bound, upper_bound,
        np.where(df['Modal_Price'] < lower_bound, lower_bound,
df['Modal_Price'])
    )

# Step 4: Feature Normalization (Min-Max Scaling for Target/Features)
if 'Yield_hg_ha' in df.columns:
    y_min = df['Yield_hg_ha'].min()
    y_max = df['Yield_hg_ha'].max()
    df['Yield_Scaled'] = (df['Yield_hg_ha'] - y_min) / (y_max - y_min)

print("Preprocessing completed successfully.")
return df

# Example Usage Demonstration
if __name__ == "__main__":
    # Simulated Raw Agricultural Data Block
    raw_data = {
        'Date': ['2025-05-01', '2025-05-01', '2025-05-02', '2025-05-03',
'2025-05-04'],
        'Region_ID': ['Reg_A', 'Reg_A', 'Reg_A', 'Reg_A', 'Reg_A'],
        'Crop_Type': ['Wheat', 'Wheat', 'Wheat', 'Wheat', 'Wheat'],
        'Precipitation_mm': [12.5, 12.5, np.nan, 0.0, 4.2],
        'Modal_Price': [2100.0, 2100.0, np.nan, 8500.0, 2150.0], # Includes
duplicate & outlier
        'Yield_hg_ha': [32000, 32000, 32500, 31000, 33000]
    }

```

```
df_raw = pd.DataFrame(raw_data)
df_clean = clean_and_preprocess_agri_data(df_raw)
```

6. Key Takeaways & Recommendations

Preparing agricultural datasets for machine learning requires a careful balance between automated statistical cleaning and domain knowledge. Blindly removing outliers can strip away vital signals corresponding to climate shocks or sudden market shifts. Moving forward, incorporating automated pipeline tools like scikit-learn Pipelines or Great Expectations can streamline validation checks and ensure data quality consistency before models are retrained in production environments.